A Course Based Project on

Adaptive Intelligent Communication System with Real-Time BER Monitoring using Simulink

Submitted to the Department of ECE

in partial fulfillment of the requirements for the completion of course

ANALOG AND DIGITAL COMMUNICATIONS LABORATORY (22PC2EC206)

BACHELOR OF TECHNOLOGY
in

ELECTRONICS AND COMMUNICATION ENGINEERING

Submitted by

N. Siddhartha	24071A04H4
P.Nithin Reddy	24071A04H7
P.Sai Manaswi	24071A04H5
U Hema Lakshmini	24071A04K1

UNDER THE SUPERVISION OF

Dr. T. Srinivas, Assistant Professor

Mrs. M. Bhagya Lakshmi, Assistant Professor

TAMASOMA JYOTIRGAMAYA
VIGNANA JYOTHI

VNRVJIET
VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
AUTONOMOUS INSTITUTION

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VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

Department of Electronics and Communication Engineering

CERTIFICATE

This is to certify that the course based project entitled "**Adaptive Intelligent Communication System with Real-Time BER Monitoring using Simulink**" is being submitted by N.Siddhartha (24071A04H4),P.Nithin_Reddy (24071A04H7),P.Sai_Manaswi(24071A04H5), U Hema Lakshmini (24071A04K1), , for the partial fulfillment of requirements for course based project of **Analog and Digital Communications Laboratory** in II year II semester of Bachelor of Technology in Electronics and Communication Engineering of VNRVJIET, Hyderabad during the academic year 2025 - 2026.

Course Coordinator	Course Coordinator	Head of the Department
Dr. T. Srinivas	Mrs. M. Bhagya Lakshmi	Dr. P. Kishore
Assistant Professor	Assistant Professor	Assoc. Professor & Head
VNRVJIET, Hyd.	VNRVJIET, Hyd.	VNRVJIET, Hyd.

DECLARATION

We do declare that the course based project report entitled "Adaptive Intelligent Communication System with Real-Time BER Monitoring Using Simulink" submitted to the Department of Electronics and Communication Engineering, Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering and Technology, Hyderabad, in partial fulfillment of the requirements for course based project of Analog and Digital Communications Laboratory in II B.Tech. II Semester for the academic year 2025 - 2026.

Place: Hyderabad Date:

	Student Name	Roll No.	Student Signature
1.	N. Siddhartha	24071A04H4	
2.	P.Nithin Reddy	24071A04H7	
3.	P.Sai Manaswi	24071A04H5	
4.	U.Hema Lakshmini	24071A04K1	

Verified by:

Dr. T. Srinivas
Assistant Professor

Mrs. M. Bhagya Lakshmi
Assistant Professor

Date of Verification:

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ABSTRACT

This project presents the design and implementation of an **adaptive intelligent communication system** that selects the most efficient digital modulation technique based on channel conditions using real-time Bit Error Rate (BER) analysis. In practical communication systems, noise significantly affects signal quality, making it important to choose a modulation scheme that performs reliably under varying noise levels.

In this work, a random binary data source is generated using a Bernoulli binary generator to represent the transmitted signal. This data is simultaneously applied to multiple modulation schemes, including BPSK, DBPSK, PAM, and FSK. Each modulated signal is then passed through an Additive White Gaussian Noise (AWGN) channel, where noise levels are varied using the Eb/N_0 parameter to simulate real-world conditions.

At the receiver side, corresponding demodulators recover the transmitted data, and the Bit Error Rate (BER) is calculated by comparing transmitted and received signals. Based on the BER values, the system evaluates and identifies the modulation technique that provides the best performance under the given noise condition.

The results show that BPSK consistently achieves the lowest BER among the considered techniques, making it the most reliable under AWGN conditions. This approach demonstrates how adaptive selection of modulation schemes can improve communication efficiency and reliability in dynamic environments.

1. INTRODUCTION

In modern communication systems, the reliable transmission of information over noisy channels is a fundamental challenge. Digital communication techniques are widely used because they offer better noise immunity, efficient bandwidth usage, and ease of processing compared to analog systems. However, the performance of any digital communication system largely depends on the choice of modulation technique, especially under varying noise conditions.

Different modulation schemes such as Binary Phase Shift Keying (BPSK), Differential BPSK (DBPSK), Pulse Amplitude Modulation (PAM), and Frequency Shift Keying (FSK) have different characteristics in terms of complexity, power efficiency, and resistance to noise. No single modulation technique is optimal for all channel conditions, which creates the need for an adaptive approach.

One of the most important metrics used to evaluate the performance of a communication system is the Bit Error Rate (BER), which measures the number of errors in the received data compared to the transmitted data. A lower BER indicates better system performance and higher reliability.

In this project, an adaptive intelligent communication system is developed that analyzes the performance of multiple modulation techniques in real time using BER. By passing signals through an Additive White Gaussian Noise (AWGN) channel and evaluating their performance at different noise levels, the system identifies and selects the most efficient modulation scheme dynamically.

This approach helps improve communication reliability and efficiency, making it suitable for real-world applications where channel conditions continuously change.

2. BLOCK DIAGRAM

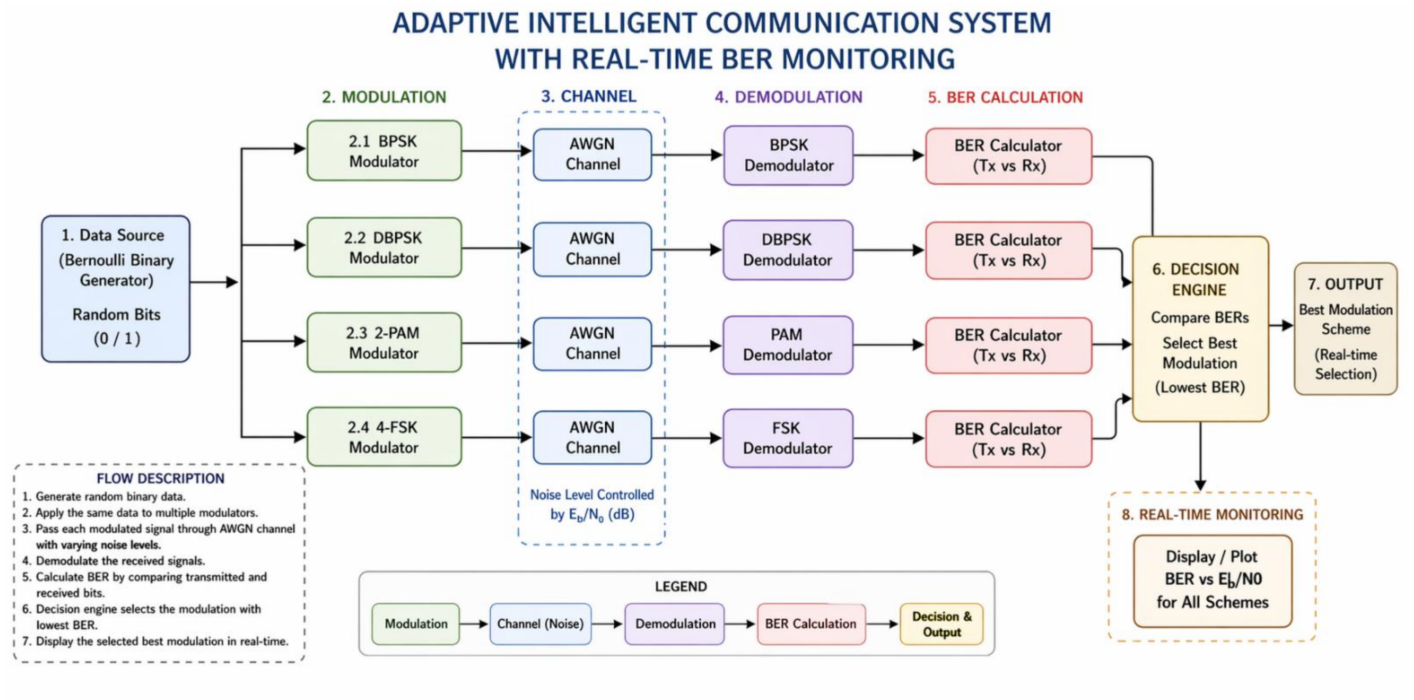

Fig 2.1: Adaptive Intelligent Communication System with Real-Time BER Monitoring using Simulink

3. WORKING

The proposed system works by intelligently selecting the most efficient modulation technique based on channel noise conditions. First, a stream of binary data is generated using a Bernoulli binary generator, which represents the transmitted signal. This same data is simultaneously applied to multiple modulation schemes, including BPSK, DBPSK, PAM, and FSK, so that all techniques can be tested under identical conditions. Each modulated signal is then passed through an Additive White Gaussian Noise (AWGN) channel, where the noise level is varied using the Eb/N_0 parameter to simulate real-world communication environments.

At the receiver side, the signals are demodulated using their respective demodulators to recover the transmitted data. The system then compares the transmitted and received bits to calculate the Bit Error Rate (BER) for each modulation scheme. Based on these BER values, a decision mechanism identifies which modulation technique performs best under the given noise condition, typically the one with the lowest BER. This process is repeated for different noise levels, allowing the system to adapt dynamically. The overall result is an intelligent communication system that continuously monitors performance and selects the most reliable modulation method in real time, ensuring efficient and robust data transmission.

4. OUTPUT

The output of the proposed system consists of the **Bit Error Rate (BER) values** obtained for each modulation technique under the same noise conditions. After transmitting the binary data through different modulation schemes (BPSK, DBPSK, PAM, and FSK) and passing them through the AWGN channel, each signal is demodulated and compared with the original transmitted data.

The system displays the BER values along with the number of transmitted and erroneous bits for each modulation scheme. These values clearly indicate how much error is introduced due to noise in the channel. By observing the outputs, we can compare the performance of all modulation techniques in real time.

From the results, it is observed that **BPSK produces the lowest BER**, indicating better reliability and resistance to noise. Other techniques like DBPSK, PAM, and FSK show comparatively higher error rates under the same conditions. Based on this comparison, the system identifies and selects the most efficient modulation scheme.

Thus, the final output of the system is:

- BER values for each modulation technique
- Comparison of performance under noise
- Selection of the best modulation method (lowest BER)

This demonstrates that the system successfully performs **real-time monitoring and intelligent decision-making**, ensuring efficient communication even in noisy environments.

Performance Evaluation Results

Modulation type	Eb/N0	2	4	6	8	10
DBPSK	0.503	0.495	0.515	0.5023	0.499	0.5127
BPSK	0.504	0.496	0.514	0.501	0.499	0.5127
PAM	0.505	0.4945	0.509	0.504	0.498	0.5125
FSK	0.497	0.4937	0.508	0.5021	0.504	0.4921

High Noise (Low Eb/N_0)

All schemes exhibit elevated BER due to signal-to-noise ratio limitations

Medium Noise

Performance differentiation becomes apparent with BPSK leading

Low Noise (High Eb/N_0)

BPSK consistently achieves lowest error rates across all conditions

5. CONCLUSION

The project successfully demonstrates an **adaptive intelligent communication system** that evaluates multiple digital modulation techniques based on real-time Bit Error Rate (BER) performance. By transmitting the same binary data through BPSK, DBPSK, PAM, and FSK schemes and analyzing their behavior under an AWGN channel, the system provides a fair and effective comparison.

From the results, it is observed that **BPSK consistently achieves the lowest BER**, making it the most reliable modulation technique under noisy conditions. While DBPSK offers the advantage of simpler receiver design without phase synchronization, it shows slightly degraded performance compared to BPSK. PAM and FSK exhibit higher error rates under the same conditions.

The system's key achievement is its ability to **monitor performance in real time and intelligently select the best modulation scheme** based on channel conditions. This approach improves communication reliability and efficiency, especially in environments where noise levels vary.

Overall, the project highlights the importance of adaptive techniques in modern communication systems and provides a strong foundation for future work in advanced modulation and real-time optimization.

6. REFERENCES

- Simon Haykin, *Communication Systems*, Wiley
- B.P. Lathi, *Modern Digital and Analog Communication Systems*, Oxford University Press
- John G. Proakis & Masoud Salehi, *Digital Communications*, McGraw-Hill
- Sklar, Bernard, *Digital Communications: Fundamentals and Applications*, Pearson
- MATLAB & Simulink Documentation – Communication Toolbox
- IEEE Papers on BER Analysis and Digital Modulation Techniques